

Answer Key & Teacher Notes

Session 1 – Area of Parallelograms · Session 2 – Area of Rhombuses

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Area of Parallelograms

TALK PROMPTS

- Why must the height make a square corner with the base you chose?
- How does the area of a parallelogram compare to a rectangle with the same base and height?

PARTNER TASK

Floor Tile and Missing Height. One partner sketches and labels the parallelogram (base, height with a square corner, slanted side); the other writes $A = b \times h$ and solves. Switch roles with Paper Strip: base 16 cm, height 7 cm, slanted side 9 cm.

WATCH FOR

- Multiplying the base by the slanted side instead of the perpendicular height.
- Labeling area in linear units (cm instead of cm^2).
- Multiplying instead of dividing to find a missing base or height.

SESSION 2 • TEACHER NOTES

Area of Rhombuses

TALK PROMPTS

- What do a rhombus and a parallelogram share, and what makes a rhombus special?
- Why does multiplying both full diagonals give too much area?

PARTNER TASK

Paper Kite. A rhombus has diagonals of 20 in. and 14 in. One partner cuts a paper model along the diagonals and rearranges the triangles into a rectangle; the other computes $(20 \times 14) \div 2 = 140 \text{ in}^2$. Switch roles with Quilt Patch.

WATCH FOR

- Multiplying the diagonals and forgetting to divide by 2.
- Using a slanted side length as the height in $A = b \times h$.
- Finding a missing diagonal without doubling the area first ($60 \div 12 = 5$).

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.G.A.1	1	2	1	1	1	2	2	3	3	1	1
S1 • Form B	6.G.A.1	1	2	1	1	1	2	2	3	3	1	1
S2 • Form A	6.G.A.1	1	1	1	1	1	2	2	3	3	1	1
S2 • Form B	6.G.A.1	1	1	1	1	1	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

parallelogram: A four-sided shape whose opposite sides are **parallel**.

base: The side you **measure** the height from.

height: The distance from the base to the opposite side, at a **square** corner.

- Area of a parallelogram: $A = \text{base} \times \text{height}$.
- The height makes a **square** corner with the base. The slanted side is **not** the height.
- Missing height or base: **divide** the area by the measure you know.

Example: base = 9 in., height = 4 in. Skip the 5 in. slanted side. $A = 9 \times 4 = 36$ Answer: The area is 36 in².

YOU TRY

1. $10 \times 6 = 60 \text{ cm}^2$
2. a. 6 in b. 9 m

EXIT TICKET

1. a. 52 in² b. 9 m
2. $11 \times 5 = 55 \text{ m}^2$
3. **B** (cm²)

PRACTICE FORM A

1. a. 126 cm² b. 72 in² c. 77 ft² d. 100 m²
2. a. $12 \times 5 = 60 \text{ ft}^2$ b. The 8 ft side is slanted; it does not make a square corner with the base, so it is not the height.
3. a. $15 \times 10 = 150 \text{ cm}^2$ b. $3 \times 150 = 450 \text{ cm}^2$
4. $90 = 18 \times h$, so $h = 90 \div 18 = 5 \text{ in.}$
5. $16 \times 7 = 112 \text{ cm}^2$ (the 9 cm side is not used)
6. $96 = 8 \times h$, so $h = 96 \div 8 = 12 \text{ in.}$
7. X: $10 \times 13 = 130 \text{ cm}^2$. Y: $26 \times 5 = 130 \text{ cm}^2$. They cover the same amount.
8. a. $15 \times 8 = 120 \text{ ft}^2$ b. $120 \div 50 = 2.4$, so 3 full cans (2 cans cover only 100 ft²) c. $30 \times 8 = 240 \text{ ft}^2$, twice the first area
9. She used the slanted side. The height must make a square corner with the base: $A = 20 \times 15 = 300 \text{ cm}^2$.
10. **C** (90 m²)
11. **A** (9 cm)

Self-check word SQUARES

PRACTICE FORM B

1. a. 80 cm² b. 52 in² c. 72 ft² d. 90 m²
2. a. $14 \times 6 = 84 \text{ ft}^2$ b. The 10 ft side is slanted; it does not make a square corner with the base, so it is not the height.
3. a. $12 \times 9 = 108 \text{ cm}^2$ b. $3 \times 108 = 324 \text{ cm}^2$
4. $84 = 12 \times h$, so $h = 84 \div 12 = 7 \text{ in.}$
5. $18 \times 8 = 144 \text{ cm}^2$ (the 11 cm side is not used)
6. $117 = 9 \times h$, so $h = 117 \div 9 = 13 \text{ in.}$
7. X: $9 \times 14 = 126 \text{ cm}^2$. Y: $18 \times 7 = 126 \text{ cm}^2$. They cover the same amount.
8. a. $20 \times 6 = 120 \text{ ft}^2$ b. $120 \div 50 = 2.4$, so 3 full cans (2 cans cover only 100 ft²) c. $40 \times 6 = 240 \text{ ft}^2$, twice the first area
9. He used the slanted side. The height must make a square corner with the base: $A = 14 \times 8 = 112 \text{ m}^2$.
10. **C** (100 ft²)
11. **A** (9 m)

Self-check word HEIGHTS

GUIDED NOTES

rhombus: A parallelogram with four sides of the **same** length.

diagonal: A segment from one corner to the **opposite** corner.

congruent: Exactly the same **size** and shape.

- A rhombus is a **parallelogram**, so $A = \text{base} \times \text{height}$ works.
- With the two diagonals, $A = d1 \times d2 \div 2$.
- Missing diagonal: **double** the area, then divide by the diagonal you know.

Example: Two diagonals are given, so use $A = d1 \times d2 \div 2$. $10 \times 8 = 80$ $80 \div 2 = 40$ Answer: The area is **40** cm^2 .

YOU TRY

1. a. 36 m^2 b. 35 cm^2
2. $2 \times 72 = 144$; $144 \div 16 = 9 \text{ in}$.

EXIT TICKET

1. a. 35 ft^2 b. 63 m^2
2. $(18 \times 10) \div 2 = 90 \text{ cm}^2$
3. **B** (12 cm)

PRACTICE FORM A

1. a. 70 in^2 b. 54 cm^2 c. 130 ft^2 d. 55 cm^2
2. A rhombus is a parallelogram: $A = 10 \times 9 = 90 \text{ cm}^2$
3. $(20 \times 14) \div 2 = 280 \div 2 = 140 \text{ in}^2$
4. $84 = 12 \times h$, so $h = 7 \text{ cm}$
5. $150 = b \times 10$, so $b = 15 \text{ cm}$
6. $(18 \times 24) \div 2 = 432 \div 2 = 216 \text{ ft}^2$
7. Double the area: $2 \times 60 = 120$. Divide: $120 \div 12 = 10 \text{ m}$.
8. a. $(30 \times 16) \div 2 = 240 \text{ in}^2$ b. $240 \div 60 = 4 \text{ strips}$ c. $(60 \times 32) \div 2 = 960 \text{ in}^2$, 4 times as large
9. She skipped dividing by 2; 160 cm^2 is the rectangle around the rhombus. $A = (10 \times 16) \div 2 = 80 \text{ cm}^2$.
10. **B** (24 ft^2)
11. **A** (All four sides are the same length.)

Self-check word RHOMBUS

PRACTICE FORM B

1. a. 54 m^2 b. 104 in^2 c. 40 m^2 d. 35 cm^2
2. A rhombus is a parallelogram: $A = 12 \times 11 = 132 \text{ cm}^2$
3. $(22 \times 16) \div 2 = 352 \div 2 = 176 \text{ in}^2$
4. $126 = 14 \times h$, so $h = 9 \text{ cm}$
5. $108 = b \times 9$, so $b = 12 \text{ cm}$
6. $(20 \times 26) \div 2 = 520 \div 2 = 260 \text{ ft}^2$
7. Double the area: $2 \times 45 = 90$. Divide: $90 \div 9 = 10 \text{ m}$.
8. a. $(24 \times 18) \div 2 = 216 \text{ in}^2$ b. $216 \div 54 = 4 \text{ strips}$ c. $(48 \times 36) \div 2 = 864 \text{ in}^2$, 4 times as large
9. She skipped dividing by 2; 108 ft^2 is the rectangle around the rhombus. $A = (12 \times 9) \div 2 = 54 \text{ ft}^2$.
10. **C** (90 in^2)
11. **C** $((d1 \times d2) \div 2)$

Self-check word POLYGON

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 • AREA OF PARALLELOGRAMS

1. $11 \times 6 = 66 \text{ cm}^2$ (the 8 cm side is not used)
2. a. 84 ft^2 b. 81 m^2 c. 80 in^2 d. $b = 63 \div 7 = 9 \text{ m}$
3. One space: $9 \times 18 = 162 \text{ ft}^2$; 10 spaces: $1,620 \text{ ft}^2$
4. $h = 108 \div 12 = 9 \text{ ft}$ (check: $12 \times 9 = 108$)
5. She multiplied the base by the slanted side (10×7). The height must be perpendicular to the base: $A = 10 \times 5 = 50 \text{ cm}^2$.

SHEET 2 • AREA OF RHOMBUSES

1. $(16 \times 10) \div 2 = 80 \text{ cm}^2$
2. a. 70 m^2 b. 63 ft^2
3. $2 \times 60 = 120$; $120 \div 10 = 12 \text{ ft}$
4. A: $(12 \times 8) \div 2 = 48 \text{ ft}^2$; B: $8 \times h = 48$, so $h = 6 \text{ ft}$
5. He multiplied the diagonals but did not divide by 2. $A = (18 \times 8) \div 2 = 72 \text{ in}^2$.
6. **C** (35 m^2)
7. **C** (90 in^2)